Assignment – 1 (30 Marks)

EE655: Computer Vision & Deep Learning

- Each question carries 6 marks.
- All questions are compulsory.
- Submit a single PDF comprising the write-up, code, and results.

Q1) Design a neural-network-based calculator that can perform addition, subtraction, multiplication, and division on any two given complex numbers. Explain the entire process. Show results for at least 10 examples.

Q2) Convert the AlexNet CNN architecture into an AlexNet FCN (Fully Convolutional Network) architecture and train it on TinyImageNet dataset. Give the diagrams of both the architectures. Provide both qualitative and quantitative results.

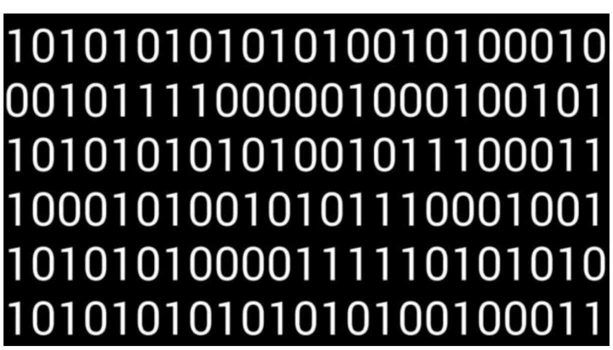
Q3) Fine-tune at least three different pre-trained networks on the following dataset:

Track Misalignment Detection dataset

In your write-up, clearly describe the following:

- Pre-trained networks used
- Training setup (hyperparameters, data augmentation, etc.)
- Performance comparison

Q4) Develop a program to extract the binary number shown in the following image:



Give a flowchart of your approach.

Q5) Implement a modified Histogram of Oriented Gradients (HoG) feature extraction algorithm from scratch, using the Robert cross edge detector for computing derivatives. Additionally, implement the LBP-based global feature (see Slide 7, Lecture 7). Extract these two features from images in the Cat and Dog dataset, concatenate them, and train an SVM classifier.